SANJANA KADAPATTI

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EXPERIENCE

Software Engineer Trainee **Aug 2025 – Aug 2026**

Cisco Systems - Bengaluru

- Architected an AI-assisted documentation workflow for **IOS-XR** AP Python code using **Python**, **AST/tokenize** metadata extraction, **LLM**-driven comment generation, and **Git/GitHub** automation, improving documentation accuracy by up to 60% while preserving runtime behavior.
- Built deterministic **Python** pipelines for intent extraction, documentation-gap analysis, comment generation, annotated file creation, and PR-ready publishing using **YAML** configuration, **CLI orchestration**, and structured metadata processing.
- Developed automated **AST/tokenize**-based code analysis to identify missing documentation across functions, classes, logging statements, and Allure steps, generating structured reports for targeted comment generation.
- Developed a **SIM topology automation framework** using **Python**, **PyYAML**, **YAML/JSON parsing**, AP/Sub-AP metadata, testcase topology variants, and CAFY POD/topology JSON inputs to generate simulator-ready configurations.
- Automated **topology inventory, validation, metadata normalization**, and **HTML/CSV coverage reporting**, reducing manual topology creation and improving consistency across simulator configurations.
- Automated end-to-end workflows for **AP/Sub-AP detection**, generated-file manifests, **Git branching**, **CDET integration**, and **GitHub PR flows**, enabling controlled and review-ready documentation changes.
- Integrated **Python scripting**, **Shell/CLI automation**, **Git/GitHub**, and **testbed workflows** to streamline **IOS-XR automation** tasks and reduce repetitive manual operations across documentation and SIM topology processes.

TECHNICAL SKILLS

- **Programming Languages**: Java, Python
- **Web Development**: HTML, CSS, JavaScript, REST APIs, GitHub API
- **Databases**: SQL, MySQL, MongoDB (NoSQL), DBMS
- **Core CS Fundamentals**: Object-Oriented Programming (OOP), Data Structures & Algorithms, Operating Systems
- **Development Tools**: Linux, Shell Scripting, Git, GitHub, GitHub Copilot, OpenAI Codex
- **Networking**: Cisco IOS-XR, CCNA-level Networking, OSI/TCP-IP, IPv4/IPv6, Subnetting, Routing, Switching, VLANs, STP, OSPF, NAT, ACLs, DHCP, DNS, Network Security, Troubleshooting
- **Network Automation**: Python Scripting, CLI Automation, YAML/JSON Parsing, CDET, Testbed Integration, PR Automation

PROJECTS

RAG-Based Document Intelligence System | *n8n, RAG, LLM, Embedding, Vector Database* **Feb 2025 – March 2025**

- Developed a Retrieval-Augmented Generation (RAG) AI agent that automatically ingests PDFs from Google Drive, generates vector embeddings, and stores them in Pinecone for semantic retrieval.
- Built an intelligent document Q&A system using n8n, enabling users to query uploaded documents through Telegram and receive context-aware, document-grounded responses.

E-Fy – A Techtainment Book (KSCST Sponsored) | *HTML5, CSS3, JavaScript, PHP, MySQL* **Jan 2024 – May 2024**

- Built a web-based event management system to streamline event scheduling, registration, and data management using HTML5, CSS3, JavaScript and MySQL.
- Implemented backend integration, input validation, and basic security practices while following SDLC concepts to develop and test the application efficiently.

EDUCATION

KLE Institute of Technology, Hubli, Karnataka **2020 - 2024**

Bachelor of Engineering in CSE *CGPA - 8.6 / 10*

ACHIEVEMENTS AND CERTIFICATIONS

- Cisco Certified Network Associate (CCNA)
- Cisco Certified AI Technical Practitioner
- Cisco Certified Network Associate Automation
- Java Basics certification - Coding Ninjas (Code 360)
- Received Cisco "**Play to Win**" Award for the project "**AI-Assisted Code Documentation Generator**"
- Published a research paper on "**AI-driven intent-aware code documentation**" in Cisco's proprietary "**Qubit**" platform
- Received **KSCST sponsorship** for E-Fy – A Techtainment Book